

SUMMARY TABLE: CORE MATHEMATICS GRADE 10

Sub-Strand 2.4: Trigonometry 1 — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 2.4: Trigonometry 1
Driving Question	DRIVING QUESTION: How can we use the angles and one side of a right-angled triangle to find any unknown height or distance — and how does this help Kenyans solve real problems in construction, surveying, and everyday life?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Noticing the Phenomenon & Building the Need: What Tool Is Missing?	Students observed that right-angled triangles appear throughout the school environment (roof trusses, staircases, ramps) and that when multiple right-angled triangles share the same acute angle but differ in size, the ratio of any two chosen sides remains nearly constant across all triangles.	Students learned that a right-angled triangle has three named sides — opposite, adjacent, and hypotenuse — defined relative to a chosen acute angle, and that the three constant side-pair ratios for a given angle are called sine ($O \div H$), cosine ($A \div H$), and tangent ($O \div A$).	Students explained that to find the height of the Kenya Power mast in Nakuru, they need the equation $\text{height} \div 40 = \tan 50^\circ$, but they cannot yet solve it because they do not know the decimal value of $\tan 50^\circ$ — identifying the precise knowledge gap that the rest of the unit will fill.
Lesson 2 Defining Sine, Cosine, and Tangent — Naming the Ratios That Unlock the Mast	Learners measured sides of right-angled triangles they drew, computed three side-pair ratios for different reference angles, contributed data to a class table on the board, and then verified their computed ratios against mathematical tables — seeing that table values match measured ratios closely, confirming that the ratios depend only on the angle.	The three trigonometric ratios have formal names and definitions: $\sin \theta = \text{opposite} \div \text{hypotenuse}$ (SOH), $\cos \theta = \text{adjacent} \div \text{hypotenuse}$ (CAH), $\tan \theta = \text{opposite} \div \text{adjacent}$ (TOA). The labels 'opposite' and 'adjacent' change when the reference angle changes. Mathematical tables and calculators give the exact value of any ratio for any acute angle, and can be used in reverse to find an angle from a known ratio.	We now have the specific numerical tool that was missing in the phenomenon. The student standing 40 m from the Nakuru mast at a 50° angle can use $\tan 50^\circ \approx 1.192$ to calculate: $\text{height} = 40 \times 1.192 \approx 47.7$ m — no ladder, no drone, no guessing. The tangent ratio is the mathematical bridge between the one known side and the unknown height.
Lesson 3 Sines and Cosines of Complementary Angles: $\sin \theta = \cos(90^\circ - \theta)$	Learners completed a table of six complementary angle pairs (e.g., 30° & 60° , 42° & 48°) using mathematical tables, recording sine and cosine values to four decimal places. They observed that $\sin \theta = \cos(90^\circ - \theta)$ and $\cos \theta = \sin(90^\circ - \theta)$ held	Learners learned that in any right-angled triangle, the sine of one acute angle equals the cosine of the other, because the side that is 'opposite' for one angle is 'adjacent' for the complementary angle. This was proved geometrically by re-labelling the	Learners applied the complementary angle relationship to: (a) rewrite trigonometric expressions (e.g., $\cos 70^\circ = \sin 20^\circ$); (b) solve for unknown angles algebraically (e.g., $\sin(2\theta + 10^\circ) = \cos(\theta + 20^\circ) \rightarrow \theta = 20^\circ$); and (c) annotate the Nakuru mast

	true across all pairs, including the mast-problem pair $\sin 40^\circ = \cos 50^\circ = 0.6428$.	same triangle from both vertices. The relationship $\sin \theta = \cos(90^\circ - \theta)$ is a structural property of right-angled triangles, not a numerical coincidence.	diagram with both the 50° elevation angle and the complementary 40° depression angle, writing the swapped sine-cosine ratios for each — showing that a Kenya Power engineer can work from either angle to find the same mast height.
Lesson 4 The Hidden Link: Discovering $\tan \theta = \sin \theta / \cos \theta$	Learners computed $\sin \theta / \cos \theta$ for six angles (20° – 70°) from their Lesson 2 data tables and found that the quotient matched their recorded $\tan \theta$ values exactly in every row, regardless of the angle chosen.	$\tan \theta = \sin \theta / \cos \theta$ for all acute angles θ . This identity follows algebraically from the definitions: $(O/H) \div (A/H) = O/A = \tan \theta$. Knowing any two of the three primary ratios allows us to determine the third.	Because $\sin \theta$ and $\cos \theta$ are both defined using the hypotenuse as denominator, dividing them cancels the hypotenuse and leaves the ratio of opposite to adjacent — which is precisely $\tan \theta$. This is why the Kenya Power engineer can recover $\tan 50^\circ$ from her notebook even without the $\tan$ button, and compute the Nakuru mast height as $\tan 50^\circ \times 40 \text{ m} \approx 47.7 \text{ m}$.
Lesson 5 Trigonometric Ratios of Special Angles (45°, 30°, 60°, 0°, and 90°)	Learners constructed an isosceles right-angled triangle (legs = 1 unit) and a bisected equilateral triangle (side = 2 units, yielding a 30-60-90 triangle) using ruler and compasses. They measured all sides and recorded the values for the 45° , 30° , and 60° triangles in a table. They also identified that as an angle approaches 0° or 90° , specific side ratios approach 0 or 1 (or become undefined), using their trig tables from Lesson 3 as evidence.	Learners derived the exact trigonometric ratios: $\sin 45^\circ = \cos 45^\circ = 1/\sqrt{2}$, $\tan 45^\circ = 1$; $\sin 30^\circ = \cos 60^\circ = 1/2$, $\sin 60^\circ = \cos 30^\circ = \sqrt{3}/2$, $\tan 30^\circ = 1/\sqrt{3}$, $\tan 60^\circ = \sqrt{3}$; $\sin 0^\circ = 0$, $\cos 0^\circ = 1$, $\tan 0^\circ = 0$; $\sin 90^\circ = 1$, $\cos 90^\circ = 0$, $\tan 90^\circ = \text{undefined}$. They applied these to find the exact height the Nakuru Kenya Power mast would have if the angle of elevation were 60° ($40 \tan 60^\circ = 40\sqrt{3} \approx 69.3 \text{ m}$) and compiled a colour-coded Special Angles Reference Card in Kenyan flag colours for use throughout the unit.	The exact values arise directly from the geometry of two special triangles — the isosceles right-angled triangle and the half-equilateral triangle — together with the definitions of sine, cosine, and tangent as ratios of sides. These exact surd values are more powerful than decimal approximations because they allow precise calculation without a calculator or table, and they reveal elegant relationships (e.g., $\sin 30^\circ = \cos 60^\circ$, confirming the complementary-angle rule from Lesson 3). For the Nakuru mast and all similar height-and-distance problems in Kenyan construction and surveying, knowing these five special angles means an engineer can solve problems in the field with nothing more than a measuring tape, a clinometer, and mental arithmetic.
Lesson 6 Solving Right-Angled Triangles Using Trigonometric Ratios	Learners observed a fully narrated worked example of the Nakuru mast problem solved using $\tan 50^\circ = h/40$, yielding $h \approx 47.7 \text{ m}$, and co-constructed the solution to the Kisumu hospital ramp problem using $\cos 12^\circ = 6/\text{hypotenuse}$, yielding $\text{hypotenuse} \approx 6.13 \text{ m}$.	Learners learned to apply the SOH-CAH-TOA decision framework to select the correct trigonometric ratio for any right-angled triangle problem, rearrange the resulting equation to isolate the unknown, evaluate using mathematical tables or a calculator, and interpret the answer in a Kenyan real-life context. They extended this to inverse tangent for finding an unknown angle (Kericho road inclination $\approx 18.4^\circ$).	Learners explained their solution steps to peers through reciprocal pair-checking (signing off on each other's working), presented full solutions to the class for two problems, and articulated a general decision rule (identify O/A/H $\rightarrow$ match to SOH/CAH/TOA $\rightarrow$ rearrange $\rightarrow$ calculate $\rightarrow$ interpret) captured in their cumulative triangle model. They connected all solutions back to the driving question about the Nakuru mast.

Lesson 7 Angles of Elevation and Depression: Solving Real Height-and-Distance Problems Across Kenya	Learners observed two carefully annotated diagrams — one showing angle of elevation (observer at base looking up at the Nakuru Power mast) and one showing angle of depression (observer at the top of Fort Jesus cliff, Mombasa, looking down at a dhow at sea level). They also observed the teacher model a live four-step solution to the anchoring phenomenon (mast height ≈ 47.7 m) and a depression problem (dhow distance ≈ 28.6 m), then observed three peer-group presentations of graded Kenya-context problems (Nairobi flagpole, Kisumu building, Mombasa cliff).	Learners learned that: (1) the angle of elevation is measured upward from the horizontal to the line of sight; (2) the angle of depression is measured downward from the horizontal to the line of sight; (3) because of alternate interior angles, the angle of elevation from A to B equals the angle of depression from B to A; (4) a four-step method (draw and label $\rightarrow$ identify known and unknown sides $\rightarrow$ select the correct trig ratio $\rightarrow$ solve algebraically) reliably solves any elevation or depression problem; (5) the tangent ratio is the primary tool when the height (opposite) and horizontal distance (adjacent) are the relevant sides; (6) real Kenyan professionals — surveyors in Rift Valley, engineers in Nairobi, mariners on the Coast — use exactly these techniques.	Learners explained, through group problem-solving and presentations, how to: select and justify the correct trigonometric ratio for elevation and depression scenarios; set up and solve equations of the form $\tan(\theta) = \text{opp/adj}$ to find unknown heights (Nairobi flagpole: ≈ 28.2 m; Kisumu building: ≈ 32.8 m) and unknown horizontal distances (Mombasa dhow: ≈ 31.1 m); connect the anchoring phenomenon (Nakuru Power mast, $h \approx 47.7$ m) back to the driving question, confirming that trigonometry provides an exact, non-invasive method for finding heights and distances using only one side and one angle of a right-angled triangle.
Lesson 8 Synthesis, Real-Life Applications, and DQB Review — Solving the Nakuru Mast at Last	Learners observed the step-by-step solution to the original Nakuru Kenya Power mast problem — the anchoring phenomenon first presented in Lesson 1 — confirming that a mast 40 m away at a 50° elevation angle is approximately 47.7 m tall, calculated using $\tan 50^\circ = h \div 40$. They also observed a two-stage Lake Victoria problem (lighthouse on a cliff near Kisumu) requiring two separate right-angled triangle diagrams linked by a common horizontal distance of 120 m, one using angle of elevation and one using angle of depression.	Learners learned that all five major concepts of the unit — trigonometric ratios (SOH-CAH-TOA), the complementary-angle relationship ($\sin \theta = \cos(90^\circ - \theta)$), the identity $\tan \theta = \sin \theta / \cos \theta$, exact values for special angles (30°, 45°, 60°), and the four-step method for solving elevation and depression problems — combine into a single, coherent mathematical tool for measuring inaccessible heights and distances. They learned that this tool is used by real Kenyan professionals including Kenya Power engineers, land surveyors in the Rift Valley, and coastguards on Lake Victoria.	The phenomenon is now fully explained: the height of the Kenya Power mast in Nakuru can be calculated without climbing it because the tangent ratio creates a precise algebraic link between the known horizontal distance (40 m), the known angle of elevation (50°), and the unknown vertical height (h). The equation $\tan 50^\circ = h \div 40$ rearranges to $h = 40 \times \tan 50^\circ \approx 47.7$ m. Trigonometry fills exactly the knowledge gap the phenomenon exposed in Lesson 1: Pythagoras relates sides to sides, but trigonometric ratios relate an angle to a ratio of sides — giving us a tool that Pythagoras alone cannot provide.

ATTRIBUTION AND LICENSING

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